

Lab 1: Requirements Engineering & UML Use-Case Modelling

Incident Escalation & On-Call Rotation Engine

Deliverable 1 — Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall ingest an incident alert, identify the active on-call engineer from the rotation schedule, and escalate the incident to the secondary engineer if the primary engineer does not acknowledge it within 5 minutes.	High	Pass: Escalation step triggered upon timeout Fail: Unacknowledged P1 incident drops without escalation	Prevents critical incidents from being missed when the primary responder is unavailable.
FR-002	Functional	The system shall allow authorized users to create, update, and view on-call rotation schedules containing primary and secondary engineers for defined shift periods.	High	Pass: The system returns the correct primary and secondary engineer for a test timestamp within that shift. Fail: The system returns an outdated or incorrect engineer.	Accurate rotation data is required for correct incident routing.
FR-003	Functional	The system shall notify the assigned on-call engineer of an incident through the configured notification channels, including SMS, email, or webhook-based delivery.	High	Pass: Notification through every enabled channel and includes the incident identifier, severity, and acknowledgment option. Fail: Any configured channel does not receive the alert.	Engineers must receive sufficient information quickly through available communication channels.

FR-004	Functional	The system shall allow an on-call engineer to acknowledge an assigned incident and shall stop further automatic escalation once a valid acknowledgment is recorded.	High	Pass: The incident status changes to acknowledged and no subsequent escalation notification is generated. Fail: Escalation continues after successful acknowledgment.	Prevents unnecessary escalation after responsibility for an incident has been accepted.
FR-005	Functional	The system shall allow the Incident Commander to record and maintain a post-mortem for a resolved incident, including its cause, impact, resolution, and follow-up actions.	Medium	Pass: Incident Commander can save all required post-mortem fields and retrieve the saved record using the incident ID. Fail: The record cannot be saved or retrieved correctly.	Supports learning, accountability, and follow-up after significant incidents.
NFR-001	Nonfunctional Performance & Security	The system shall initiate alert dispatch through webhook, SMS, and email within 3 seconds of incident ingestion.	High	Pass: The measured time from alert ingestion to dispatch initiation is ≤ 3 seconds for every tested alert. Fail: Any tested alert exceeds the 3-second limit.	Rapid notification is essential for reducing response time during critical incidents.
NFR-002	Nonfunctional Security	The system shall restrict modification of on-call schedules and incident post-mortem records to authorized users according to their assigned roles.	High	Pass: Authorized roles can perform permitted modifications while unauthorized users receive an access-denied response for every protected operation. Fail: An unauthorized user can modify a protected record.	Prevents unauthorized changes to operational schedules and incident records.

